

Children’s acquisition of the meaning of disjunction

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Abstract

The question of how young children interpret disjunctive words like English “or” has become a topic of increased attention in the recent developmental semantic/pragmatic literature. We present a thematic overview of the past half century of research on children’s acquisition of the meaning of disjunction, from the Piagetian era in the 1940s to the most recent experimental pragmatics studies of disjunction. Along the way, we highlight ways in which children differ from adults in their interpretations of disjunction, as well as ongoing debates about the sources of these interpretations, with the hope that a historical perspective may shed light on these recent debates.

1 Introduction

The question of how young children interpret disjunctive words like English “or” has become a topic of increased attention in the recent developmental semantic/pragmatic literature. In this article, we present a thematic overview of the past half century of research on children’s acquisition of the meaning of disjunction (for an overview of disjunction itself, see Aloni 2025).

The literal meaning of disjunction is generally assumed to correspond to logical disjunction, which has an inclusive meaning: a sentence such as “Lilo will bring pizza or sushi” means that Lilo will bring pizza or sushi, or both, and is thus true as long as at least one disjunct is true. In everyday conversation, however, adults tend to strengthen this interpretation to an exclusive one, namely *Lilo will bring pizza or sushi, but not both*. This *not both* inference is generally treated as a scalar implicature (specifically, an *exclusivity* implicature) arising from the competition between disjunction and the stronger conjunctive alternative “and”. One reason is that if the stronger conjunctive alternative were true, the speaker would have chosen to utter it rather than the logically weaker disjunction; this leads us to infer that the speaker does not believe the conjunction to be true, which can then be strengthened to the inference that the conjunction must be false, hence the *not both* meaning (Grice 1975; Horn 1972; van Rooij and Schulz 2004; Sauerland 2004; Spector 2007).

While it is relatively uncontroversial that adults can (and often prefer to) interpret disjunction exclusively, the story appears to be quite different for children. Instead of the exclusive interpretation, children reportedly favor inclusive or conjunctive readings of disjunction. The finding that children sometimes interpret disjunction conjunctively has led to much recent debate about the source of this behavior, with some accounts attributing it to a scalar implicature (Singh et al. 2016), a semantic ambiguity between disjunction and conjunction (Sauerland and Yatsushiro 2018), or a mere experimental artifact (Skordos et al. 2020; Huang and Crain 2020).

In this review, we aim to shed new light on these recent debates by bringing to the forefront what might be considered forgotten studies of children’s acquisition of disjunction. We will present these studies in the perspective of the research questions that motivated them, and see that evidence of conjunctive behavior surfaces in the literature as far back as the 1950s. After introducing the current debate about the source of conjunctive behaviors in more detail, we will discuss how these older results can inform this debate, and the blind spots they reveal in current research.

In Table 1, we list the studies included in this review, with information about sample sizes, age ranges, languages tested, and task types. For reasons of space, we have excluded studies that examine disjunction from a purely logical perspective using non-verbal tasks (e.g., Youniss and Furth 1964, 1967; Snow and Rabinovitch 1969; Furth et al. 1970; Siegel and Forbes 1970; Bourne and O’Banion 1971; Ciborowski and Cole 1972) and studies focusing on contradictions and tautologies (e.g., Osherson and Markman 1974). Note that there are still ongoing debates regarding children’s disjunctive reasoning abilities (see Leahy and Carey 2020 for a review), and how these connect to their linguistic abilities (Grigoroglou and Ganea, 2022; Saponaro et al., 2026), but the focus tends to be on children younger than those featured in studies of linguistic disjunction.

Previewing the central themes and research questions underlying this review, Table 2 presents a condensed summary of the main conclusions and open questions.

Study		<i>N</i>	Age range	Language	Task
Nitta and Nagano (1966)		679	5–18*	Japanese	GX
Neimark and Slotnick (1970)		223	8–14*	English	GX
Neimark (1970)		162	14–18*	English	GX, FP
Suppes and Feldman (1971)	Exp 1	64	4;5–6;7	English	GX
	Exp 2	112	4;6–6;0	English	GX
Paris (1973)		160	7–15*	English	GX, TV
Johansson and Sjölin (1975)		60	2;0–7;6	Swedish	GX, FP
Johansson (1977)		60	6–12	Swedish	GX, o
Hatano and Suga (1977)	Exp 1	105	9;1–12;0	Japanese	GX, FP
	Exp 2	145	8;10–11;9	Japanese	GX, FP
Sternberg (1979)		188	7–18	English	I
Braine and Romain (1981)		66	5–10	English	GX, TV, I
Winer and Falkner (1984)	Exp 3	226	6–10	English	o
Kwock (1984)	Exp 1	120	8;2–13;6	English	o
	Exp 2	94	8;6–13;3	English	o
	Exp 3	92	8;6–13;0	English	o
Winer et al. (1987)	Exp 1	48	7;1–11;7	English	o
	Exp 2	72	9;5–16;5	English	o
Crain et al. (2000)		13	3;5–6;1	English	TV
Gualmini et al. (2000b)	Exp 1	16	4;1–6;1	English	TV
	Exp 2	13	4;1–6;1	English	TV
Gualmini et al. (2000a)		14	3;2–5;9	English	TV
Chierchia, Crain, Guasti, Gualmini and Meroni (2001)	Exp 1	15	3;7–6;3	English	TV
	Exp 2	15	3;5–6;2	English	TV
	Exp 3	15	3;2–6;0	English	o
Chierchia et al. (2004)		9	5;1–6;0	Italian	TV
Goro and Akiba (2004)		60	3;7–6;3	Japanese	TV
Jing et al. (2005)		21	3–6	Mandarin	TV
Verbuk (2006)		21	3;11–6;10	Russian	o
Notley et al. (2012a)		22	4;2–5;2	English	TV
Notley et al. (2012b)		24	3;4–5;1	English	TV
		20	4;6–5;4	Mandarin	TV
Su (2014)	Exp 1	32	2;6–5;0	Mandarin	TV
	Exp 2	39	3;11–5;5	Mandarin	TV
	Exp 3	29	3;11–5;6	Mandarin	TV
Geçkin et al. (2016)		31	4;1–5;11	Turkish	TV
Geçkin et al. (2018)		24	4;1–5;8	German	TV
Tieu et al. (2016)	Exp 1	22	3;7–4;9	Mandarin	TV
	Exp 2	15	4;1–6;8	English	TV
Hochstein et al. (2016)		43	4;0–5;11	English	o
Singh et al. (2016)		56	3;9–6;4	English	TV
Tieu et al. (2017)		28	3;7–6;6	French	TV
		18	4;7–6;6	Japanese	TV
Sauerland and Yatsushiro (2018)		64	4;2–8;6	German	TV
Pagliarini et al. (2018a)		20	4;1–5;8	English	TV
Pagliarini et al. (2018b)	Exp 1&2	48	3;5–6;0	Italian	TV
Skordos et al. (2020)		126	4;0–5;11	English	TV
Huang and Crain (2020)	Exp 1&3	22	4;10–5;8	Mandarin	TV
	Exp 2	23	3;8–5;11	Mandarin	TV
Jasbi and Frank (2021)	Exp 2	42	3;1–5;2	English	TV
	Exp 3	50	3;6–5;9	English	TV
Shimada and Goro (2021)		45	4;7–6;1	Japanese	TV
Pagliarini et al. (2022)		20	4;2–6;1	Dutch	TV
		26	4;4–5;10	French	TV
		27	4;1–6;5	Hungarian	TV
Cochard et al. (2023)		96	3;6–8;5	French	TV
Cochard et al. (2025)		57	3;11–6;9	French	TV
Bleotu et al. (2024)		69	4;11–5;11	Romanian	TV
Bleotu et al. (2025b)		257	5–6	Romanian	TV
Bleotu et al. (2025c)		168	4–6	Romanian	TV
Bhaumik (2025)		74	4–5	Bengali	o
		48	8–9	Bengali	o
Bhaumik (2026)		40	4;0–5;10	Bengali	o
		33	8;0–10;8	Bengali	o
Zhu and Chen (2026)	Exp 1	30	4;9–6;6	Mandarin	TV
	Exp 2	30	4;6–6;9	Mandarin	o
Bhaumik and Jasbi (2026)		41	4;1–5;11	Bengali	GX

Table 1: Studies included in this review, including sample sizes (excluding adult controls), age ranges, languages tested, and task used. *For papers that only report school grades, age range is estimated. The tasks are coded as follows: GX = Give-X, TV = truth-value, FP = free production, I = inference, o = other

Theme	What we learn from the literature	Outstanding questions
How children interpret disjunction	• Unlike adults, children rarely interpret disjunction exclusively. • Children occasionally treat disjunction as conjunction. • The rate of conjunctive behavior varies greatly between studies and is sensitive to subtle aspects of the experimental design.	• When and how do children transition from the conjunctive or inclusive interpretation to adults' primarily exclusive interpretation? • Which factors determine the rate of conjunctive behavior?
The source of children's conjunctive behavior	• One account treats children's conjunctive behavior as the result of an implicature (Singh et al., 2016). • An alternative view is that conjunctive behavior simply reflects experimental artifacts (Skordos et al., 2020, a.o.).	• How can we determine whether conjunctive behavior reflects a genuine interpretation or various experimental artifacts? • Can the implicature account explain variation without overfitting the data?
Natural occurrence of disjunction words	• 'or' never appears as a propositional disjunction in children's input, instead arising mostly in alternative questions, and otherwise in free choice constructions. • Children's production of 'or' reflects the input. • Children seem to understand alternative questions much better than disjunctive statements.	• How can children acquire the logical meaning of disjunction from its use in questions and under modals? • Could properties of the input explain children's conjunctive behavior in plain declaratives?

Table 2: Summary of main findings and open questions

2 The Piagetian era

The earliest studies on the acquisition of disjunction were part of a research project initiated by Piaget in the 1940s. Piaget was primarily interested in the acquisition of logical operators, rather than their linguistic realizations, with the latter serving as a tool to test children's understanding of the underlying operations. Within Piaget's framework, conjunction is acquired in an earlier stage of development than disjunction. Specifically, conjunction is acquired during the "concrete operational" stage (starting around age 7), during which children are able to reason about concrete objects and familiar situations. Disjunction, on the other hand, is acquired during the "formal operational" stage (age 11 and into adolescence), characterized by the emergence of more abstract and hypothetical reasoning. Given this, few studies of disjunction from this period examined preschoolers, and the most common methodology was a pen-and-paper test administered to a whole class of children at once.

Inhelder and Piaget (1959) report qualitative studies with a small number of child participants, finding that children often treated disjunction as a conjunction. With Nitta and Nagano (1966) came the first large-scale study. These authors conducted 10 different surveys with Japanese children from kindergarten age to high school, testing various modalities (verbal, visual, with pictures of either concrete objects or symbols), and using affirmative and negative prompts such as "Circle all pictures which are a bird or black" and "Cross all pictures which are not a bird or black". In the affirmative case, conjunction was usually interpreted correctly, while disjunction was often interpreted conjunctively, a finding the authors took to be consistent with Piaget's theory. Notably, conjunction was more difficult with negation, while the opposite was true for disjunction.

Nitta and Nagano's results were subsequently replicated by Neimark and Slotnick (1970), who tested American children and college students. Neimark (1970) then extended the earlier findings to highschoolers, and compared " A or B " with " A or B or both", finding no clear difference between the two expressions.

With a focus on younger children aged 4 to 6 years, Suppes and Feldman (1971) explored different linguistic ways to express logical connectives using a "Give X" task. In their Experiment 1, only two of the commands used to test set union involved a disjunction (others relied on conjunction, e.g., "Give me the red things and the square things"); see (1). In Experiment 2, the authors used more systematic stimuli, the most relevant of which are listed in (2).

- (1) Commands with disjunction in Suppes and Feldman (1971, Exp. 1):
 - a. Give me the green things, or, the round things.
 - b. Give me the things that are green or square.
- (2) Commands used in Suppes and Feldman (1971, Exp. 2):
 - a. Give me the things that are round or green.
 - b. Give me the red or star things.
 - c. Give me the stars or the green things.

In response to (1a), most children gave only the green things or only the round things, as expected. In response to (1b), however, most children gave the intersection (i.e. all and only the green squares), and only 3% gave the union (i.e. all things that were green, squares, or both). In Experiment 2, participants treated the three commands in (2) similarly, with around 30% of responses reflecting intersection, 20% the first-mentioned set, and 15% the second-mentioned set; no other clear patterns emerged.

Paris (1973) was the first to explicitly shift the discussion from logical operators to linguistic connectives. He tested both “or” and “either...or”. Importantly, his was also the first study to use declarative sentences and truth-value judgments rather than a “Give X” task with imperatives. The results suggested that children and teenagers were mostly conjunctive, and only college students showed an exclusive interpretation of disjunction. The complex disjunction “either...or” generated fewer conjunctive and more exclusive interpretations.

To summarize, this period of research contributed a number of large-sample studies conducted with school-aged children, and some preschoolers. Most of the studies concluded that disjunction is acquired during adolescence, and usually not before high school. While the research program initially focused on logical operators, the discussion of linguistic factors became more central over time. On the theoretical side, the dominant view remained close to that of the Piagetian view of distinct concrete and formal operational stages, although some challenging data were beginning to emerge. Of note, McLaughlin (1963) offered an alternative formalization of the Piagetian theory, while Neimark and Slotnick (1970) discussed certain concrete mechanisms to explain children’s behavior.

3 The emergence of linguistic factors

Studies conducted from the mid-1970s were characterized by a more careful consideration of linguistic factors and task effects, with the emergence of at least some informal reflection on the pragmatics of the tasks used to study disjunction, something already prefaced in Paris (1973).

Against the background of the semantics of “and” and “or” proposed in Dik (1968), Johansson and Sjölin (1975) tested children aged 2 to 8 years on a free production task in which they had to combine pictures and describe them, and two modified Give-X comprehension tasks where children responded to imperative prompts (e.g., “Put up the car or the bicycle!”) by putting pictures on a board; the prompts were either preceded by a context story or presented without context. The authors found that in production, the conjunction “and” was produced much more frequently than the disjunction “or”, and was used primarily for enumeration (i.e., mereological sums, see fn. 1). “Or” mainly occurred with modals with ‘free choice’ effects (which we discuss in §4) and alternative questions. Notably, children did not use “or” to express set union.

In the two comprehension tasks, the authors found that children understood both “and” and “or” from age 4, with no difference in performance between the two connectives. Prior to age 4, the most common error was to give all of the pictures, even those corresponding to unmentioned

objects, but there was no indication of any conjunctive interpretation of disjunction. The authors also found that having a prior context story helped children in the Give-X task. Johansson and Sjölin concluded that children have acquired an adult-like semantics for both “or” and “and” by age 4 and that context helps them retrieve the intended interpretation.

In a follow-up, Johansson (1977) asked participants to define the meanings of the words “or” and “and”; participants then completed a Give-X task testing both narrow and wide-scope variants of both connectives. Participants’ definitions for “and” reportedly referred to combinations, while their definitions for “or” usually made reference to alternatives. Children’s performance on the Give-X task correlated with the level of abstraction of the definitions they provided in the first task (though age does not appear to have been controlled for). As expected, participants mostly interpreted the narrow scope conjunction (“Circle all figures that are blue and square”) as set intersection, giving only figures that were both blue and square, but some younger children also gave set-union responses, offering figures that were only blue, only square, or both. The wide scope conjunction (“Circle all figures that are blue, and all that are square”), on the other hand, only elicited set-union responses. For disjunction, most participants gave exclusive responses for both narrow and wide scope connectives, with the exception of some older participants who gave inclusive responses to the narrow scope variant.

Hatano and Suga (1977), seemingly unaware of Johansson’s work, reinterpreted children’s conjunctive behavior as follows: in natural settings, disjunction usually involves non-overlapping classes, and children might generalize that union presupposes no overlap. By contraposition, when they encounter two overlapping classes, they conclude that what is expressed is not union, but intersection. In other words, children rely on a heuristic instead of the lexical semantics of the connective to figure out what logical operation is relevant in the context. From this, Hatano and Suga predicted that disjunctions of non-overlapping classes (“black or red”, in contrast to “black or square”) should be easier for children — something already noted in Nitta and Nagano (1966), and that in production, children should not use a disjunction “A or B” to express the union of two overlapping classes (e.g., black or square).

Hatano and Suga replicated the finding that children struggle with disjunctions of overlapping classes (18% and 35% correct responses in grades 4 and 6, respectively), but with non-overlapping classes, performance was much higher (42% and 75%, respectively). In a production task, they found that virtually no child used the disjunction “ka” to describe a union of two overlapping classes, nor the direct mereological sum “A to B”,¹ but instead overwhelmingly relied on more complex sums of non-overlapping classes, such as “the black ones and the red-but-square ones”, in line with the behavior of American high-schoolers in Neimark (1970). In a second experiment using the same task with additional con-

¹In logical terms, a mereological sum refers to a combination of entities (e.g., “Ann and Bill”), while a conjunction is a combination of propositions or predicates (e.g., “John danced and sang”) (see Link 1983). English, unlike Japanese, does not distinguish between the two concepts and expresses both with ‘and’, but the difference becomes clear with collective predicates: “Ann and Bill met” does not mean “Ann met and Bill met”.

trols, and a new task where they tested mereological sums similar to what the children produced in Experiment 1, Hatano and Suga replicated the results of previous experiments, and found that the complex sums helped with difficult unions $A \cup \bar{B}$ and $\bar{A} \cup \bar{B}$, but reduced performance on the simpler $A \cup B$ compared to a standard disjunction.

In order to determine whether children’s difficulty with disjunctive statements in reasoning tasks stemmed from the encoding of the disjunctive premise (which would point towards a non-adult semantics) or from the combination of the different premises (which would indicate limitations in logical reasoning, as argued by Piaget), Sternberg (1979) used a method of task decomposition: two tasks were run between-subjects (inference from a single premise, and from the combination of two premises), where success on the second task should presuppose success on the first. Test-retest reliability was estimated with a second session. All age groups were highly consistent between sessions, with the exception of second graders, whose performance fluctuated. Having found no difference in performance between the two tasks, Sternberg concluded that children’s difficulty stemmed primarily from linguistic encoding, rather than logical reasoning. He also examined the top three response patterns for each age group. For disjunction, he found a transition from an inclusive to an exclusive interpretation, with no evidence of a conjunctive interpretation.

Braine and Romain (1981) criticized previous studies on the grounds that these had focused on the truth conditions children assigned to disjunction instead of their reasoning abilities. In their view, the core meaning of a connective is given by the inferences it permits, while truth tables are derived from these rules. They also pointed out a number of methodological issues, such as unnatural sounding sentences and scope ambiguities (already discussed by Johansson and Sjölin 1975). In their own study, Braine and Romain tested children on four different tasks. Task 1 tested children’s understanding of instructions with wide and narrow scope disjunctions. Task 2 tested contradictions and tautologies. Task 3 was a truth-value judgment task, and Task 4 tested children’s ability to reason with disjunctive premises.

In the first task, all children and most adults gave only one set of objects in response to both commands, accessing a wide-scope interpretation of disjunction in both cases (the conflict with Johansson and Sjölin 1975 is attributed to minor differences in the design). In Task 3, children interpreted disjunction inclusively, conjunctively, or exhibited a mix of the two interpretations. The rate of conjunctive interpretations was highest in the youngest age group (5–6 years), for whom conjunctive interpretations were more frequent than inclusive interpretations; but conjunctive readings remained relatively frequent in the 7–8 and 9–10 age groups. Adults, on the other hand, were split between inclusive and exclusive interpretations. Finally, in Task 4, all participants correctly answered all problems, with the youngest children giving 73%–91% correct answers to the different schema. Based on these findings, Braine and Romain concluded that children master the inferential properties of disjunction long before showing adult-like truth-value judgments.

In two studies conducted with adults and one with children aged 6 to 10 years, Winer and Falkner (1984) attempted to prime inclusive and

exclusive interpretations. Participants were exposed to alternative questions such as “Is this an animal or a dog?” or “Is this an animal or a book?”, asked about either dogs or non-dog animals and received positive or negative feedback. Children were then tested on Piaget’s class inclusion task, which required them to say whether there were more dogs or more animals in a picture. Winer and Falkner found no effect of the type of pre-test question in 6 and 8-year-olds, who almost always answered that there were more dogs than animals. In contrast, the question manipulation had an effect on 10-year-olds, leading the authors to conclude that children’s difficulties with the class inclusion task were primarily linguistic in nature. Follow-up studies reported in Kwock (1984) and Winer et al. (1987) reached similar conclusions and, notably, were also the first to cite theoretical work in pragmatics (Austin, 1962; Grice, 1975; Searle, 1969).

Aside from Winer and colleagues’ work, interest in the acquisition of disjunction faded in the 1980s. Few other studies involved experimental work on children’s acquisition of disjunction, and these fall outside the scope of this review (they focus on the logical operation instead of linguistic disjunction, see Aubert 1998; involve second language acquisition, see Carro-Kowalczyk 1982; or focus on children’s understanding of tautologies and contradictions, see Peterson and Marrie 1988).

4 The advent of experimental pragmatics

While interest in the acquisition of disjunction waned in the 1980s, the late 1990s and early 2000s saw the advent of experimental pragmatics, and with it, a renewed interest in children’s interpretation of disjunction. Studies in this period investigated whether children have an inclusive semantics for “or” in contexts where adults do, and how their pragmatic abilities differ from those of adults. From a methodological perspective, the main innovation that separated this wave of research from previous studies was the introduction of the modern Truth Value Judgment Task (Crain and McKee 1985; Crain and Thornton 2000) (hereafter TVJT), which uses a puppet’s utterances as stimuli, making it easier for children to reject an infelicitous sentence. By contrast, most previous studies used a Give-X task,² or required children to judge an adult experimenter’s utterance (Paris 1973).³

Gualmini et al. (2000b) investigated how children interpret disjunction in modal sentences. Experiment 1 tested sentences in which the disjunction scoped above the modal, such as “Batman or Spiderman can finish a puzzle.” In contexts in which both disjuncts were true, children reportedly accepted the test sentences on 85% of the trials; when only one

²Among other limitations, Jasbi and Frank (2021) point out that this task may have overestimated exclusive interpretations, because even if a child reaches an inclusive interpretation (“Give A, B, or both”), giving only one set of items (usually A) is the easiest way to satisfy the command.

³As a note of caution, early studies in experimental pragmatics involved relatively small sample sizes (usually 15 to 20 children, sometimes even fewer). Fortunately, most of the results of these earlier studies were later confirmed by better-powered studies.

disjunct was true, the rate of *yes*-responses dropped to 67%. Crucially, children rejected conjunctive controls in the latter scenarios 85% of the time, leading the authors to conclude that children can successfully distinguish disjunction and conjunction, contra Paris (1973). Adults derived free choice inferences (*Batman can finish a puzzle and Spiderman can finish a puzzle*) from the modal disjunctive statements, uniformly rejecting when only one disjunct was possible. The authors conducted a second experiment in which neither puppet could complete the action alone, but were able to do it together. In such a context, children reportedly accepted the disjunctive sentences only 50% of the time (while accepting conjunction at a rate of 96%). Adults, on the other hand, uniformly rejected the disjunctive targets.

In another study, Gualmini et al. (2000a) observed that when a puppet said “If a giraffe or a penguin is on the stage, then I get a coin”, children understood that the puppet would get a coin whether one animal (86%) or both animals (98%) were on the stage — consistent with an inclusive reading of disjunction. Children also correctly rejected conjunctive controls in such contexts at a rate of 76%.

The same group of researchers also investigated disjunction in negative sentences, reporting inclusive interpretations of sentences such as “The frog will not choose the hot dog or the melon” (Crain et al. 2000). Children uniformly rejected such statements in contexts in which the frog chose only one object, and likewise rejected the sentences when the frog chose both of the mentioned objects.

More evidence that children could indeed distinguish disjunction and conjunction was reported in Chierchia et al. (2001). The first two experiments employed a TVJT to investigate how children interpret disjunction in upward- and downward-entailing environments, specifically in the nuclear scope and the restrictor of the universal quantifier “every”. In their first experiment, the authors tested disjunction in the restrictor of the universal quantifier (e.g., “Every dwarf who chose a banana or a strawberry received a jewel”, in a context in which all of the dwarves in the story picked both a banana and a strawberry); in their second experiment, they assessed the interpretation of disjunction in the nuclear scope of the universal quantifier (e.g., “Every boy chose a skate-board or a bike” in a context in which every boy chose both a skate-board and a bike) (see also Boster and Crain 1993). While adult controls showed a clear distinction between the two kinds of environments, computing exclusivity implicatures only when the disjunction appeared in the nuclear scope of the quantifier, children as a group made less of a distinction between the two environments (though the authors identified two sub-groups of children, one that computed exclusivity implicatures like adults, and one that interpreted disjunction inclusively). Overall, the results of the two experiments suggested children derived fewer exclusivity implicatures from disjunction than adults.

In a third experiment, the authors used a Felicity Judgment Task in which two puppets described the given scenarios, one of whom used the disjunction (e.g., “Every farmer cleaned a horse or a rabbit”) and one of whom used conjunction (e.g., “Every farmer cleaned a horse and a rabbit”). On this task, children performed well, favoring the puppet who

uttered the conjunctive description in the critical conjunctive contexts. Based on these findings, the authors concluded that when the relevant alternatives are made readily available, children can successfully compute exclusivity implicatures.⁴ In subsequent work, Chierchia et al. (2004) corroborated the previous findings with Italian children.

Research investigating disjunction in negative environments provided further arguments that children do not simply confuse disjunction and conjunction. Goro and Akiba (2004) showed that Japanese-acquiring children tend to interpret disjunction inclusively, within the scope of negation (giving rise to a ‘neither...nor’ interpretation). This is despite the fact that Japanese adults treat the disjunction as a positive polarity item, scoping it above negation. Goro and Akiba took their findings to be consistent with Crain et al.’s (1994) Semantic Subset Principle, with children favoring the stronger reading of scopally ambiguous sentences for which one reading entails the other.⁵ This conclusion was subsequently supported by findings from several languages (Jing et al., 2005; Verbuk, 2006; Geçkin et al., 2016; Pagliarini et al., 2018b), though see Pagliarini et al. (2022) for some counterpoints. Additional studies show that children interpret disjunction inclusively in other downward-entailing contexts (speaking against any confusion between disjunction and conjunction) (Chierchia et al. 2001; Gualmini et al. 2000b; Chierchia et al. 2004; Crain 2008; Crain and Khelentzos 2010; Notley et al. 2012a,b). Such findings dovetail with Nitta and Nagano (1966)’s much earlier work showing that children performed much better when asked to cross out items which were “not A or B”, than when asked to circle those that were “A or B”.

It is worth noting that within the same decade, studies were conducted investigating children’s abilities with a range of scalar implicatures beyond the exclusivity implicature of disjunction, including the implicatures of the existential quantifier “some”, the modal verb “might” (Noveck 2001), numerals (Barner and Bachrach 2010), and verbs like “start” (Papafragou and Musolino 2003). The general consensus in the early 2000s was that children did not compute scalar implicatures to the same degree as adults. One explanation linked children’s behavior to the nature of the alternatives required to compute the relevant inferences; specifically, children were posited to have difficulty accessing stronger *lexical* alternatives. Evidence for this hypothesis came from findings that children could successfully derive implicatures in cases where the required alternatives were more easily accessible, for example, in the case of ad hoc implicatures, which involve contextual rather than lexical alternatives (see Barner and Bachrach 2010; Barner et al. 2011; Stiller et al. 2015; Tieu et al. 2016).⁶

Disjunctions are particularly relevant for this debate because they in-

⁴Note, however, that selecting the stronger alternative in such as task does not necessarily reflect the calculation of an implicature; it could simply reflect recognition of the relative strength of the two alternatives.

⁵The Semantic Subset Principle is rooted in considerations of learnability. If children were to start out hypothesizing the logically weaker superset reading, no negative evidence could lead them to revise their hypothesis, since any evidence consistent with the stronger subset reading would also be compatible with the superset reading.

⁶See Katsos and Bishop (2011) for an alternative explanation, which posits that children are simply more *pragmatically tolerant* than adults.

volve multiple types of alternatives: on the one hand, the connective “or” competes with its lexical alternative “and”, resulting in the exclusive interpretation (Grice 1975; Horn 1972). On the other hand, the individual disjuncts can themselves be relevant contextual alternatives, and in fact are responsible for *ignorance inferences* (the speaker does not know which of the disjuncts may be true; see Sauerland 2004 and Katzir 2007).

Further arguments for the alternatives-based hypothesis can be found in developmental studies of *free choice* (FC) inferences. The sentence “KungFu Panda is allowed to push the orange car or the green car” gives rise to the inference that *KungFu Panda is allowed to push the orange car and he is allowed to push the green car*. On certain accounts, this conjunctive inference is derived as a quantity implicature (Kratzer and Shimoyama, 2002; Alonso-Ovalle, 2006; Fox, 2007; Franke, 2011; Bar-Lev and Fox, 2020; Champollion et al., 2026); crucially, the required alternatives do not involve lexical replacement, but rather sub-constituents of the disjunctive sentences themselves, namely each of the individual disjuncts, strengthened with its own implicatures: *KungFu Panda is only allowed to push the orange car*, *KungFu Panda is only allowed to push the green car*. Negating these alternatives leads precisely to the FC inference.⁷

Tieu et al. (2016) presented evidence from a TVJT that 4- and 5-year-old Mandarin-acquiring children could successfully compute free choice inferences from modal disjunctive sentences, rejecting sentences such as “KungFu Panda is allowed to push the orange car or the green car” in contexts in which he was only allowed to push one of the colored cars. The high rejection rate observed in the critical trials (91%, 80/88 trials) was in stark contrast to the same children’s performance on scalar implicatures involving lexical alternatives (the modal verb *keyi* ‘may’ and the plain disjunction *huozhe* ‘or’), for which the rejection rate was only 18% (16/88 trials). Tieu et al. took these findings to support a version of the alternatives-based hypothesis (Barner et al. 2011), arguing that children struggle when the required alternatives are lexical in nature, but perform well when the alternatives are contextually/structurally provided (for a parallel asymmetry in adults’ processing of implicatures and free choice, see Chemla and Bott 2012).

Hochstein et al. (2016) further pointed out that under the alternatives-based account, children should nevertheless be able to compute ignorance inferences (as to which disjunct is true), since these do not rely on lexical alternatives. The authors tested 4- and 5-year-olds on two tasks in which children had to identify which of two puppets had uttered a target sentence. In the task examining ignorance inferences, one of the puppets was blindfolded, while the other could clearly observe the whole scene. In the scalar implicature task, both puppets saw the action, but one was described as “always [saying] things that are just right” and the other as “always [saying] things that are a little weird or silly.” As expected, both 4- and 5-year-olds failed to derive scalar implicatures, but 5-year-olds were able to derive ignorance inferences.

The insights provided by experimental pragmatics studies shifted the

⁷There are also accounts which do not treat FC as an implicature (Zimmerman 2000; Goldstein 2019; Aloni 2022, a.o.)

perspective on the previous findings on disjunction, without necessarily contradicting them. The Piagetian view had found support in the observation that at least some children were not adult-like with disjunction until well after they had already mastered conjunction. The implicature studies of the early 2000s, however, focused on the observation that, as a group, even young children distinguished disjunction from conjunction (which only requires that some of them do not interpret disjunction as conjunction). This subtle difference seems to have been missed in Chierchia et al.’s (2004) comments on Paris (1973): “One result from previous research is that second graders do not distinguish *and* from *or* (Paris, 1973). However, in an experiment we carried out, it was established that children access the full range of truth-conditions licensed by the semantic component and distinguish between *or* and *and*.” Paris in fact did not argue that children as a group did not distinguish “and” from “or”, only that a significant portion of them did not.

The fact that children were able to distinguish “or” from “and” in downward-entailing environments reinforced the idea that the conjunctive behavior observed in older studies was primarily an experimental artifact. Consequently, the studies described in this section typically didn’t test whether children would reject a disjunction when only one disjunct was true. Without this condition, these studies would not be able to distinguish inclusive from conjunctive interpretations.

5 Revisiting conjunctive readings from a pragmatic perspective

A study by Singh et al. (2016) brought back to the foreground children’s conjunctive interpretations of disjunction, and provided a novel account of this non-adult-like behavior. Singh et al. used a TVJT to test English-speaking children’s interpretation of disjunctive sentences such as “The boy is holding an apple or a banana” and “Every boy is holding an apple or a banana.” Their child participants tended to reject the disjunctive sentences in contexts in which only one disjunct was true, more than they did when both disjuncts were true — consistent with a conjunctive interpretation of disjunction, rather than an inclusive or exclusive one. Acknowledging that earlier work had also reported evidence of conjunctive readings (e.g., Paris, 1973; Braine and Romain, 1981), Singh et al. proposed a novel explanation for children’s conjunctive behavior, tying it to the way that adults compute free choice inferences from modal disjunctive statements. On their account, children differ from adults in the set of alternatives that they access for disjunction: while adults access the conjunctive alternative, leading to the exclusivity implicature, children access only the individual disjunct alternatives (each enriched with its own implicatures), leading to a conjunctive inference — akin to how adults compute FC inferences from modal disjunctive statements. The exact derivation is not directly relevant to the debate, but interested readers will find an informal presentation in appendix A.

It is worth noting that the materials and experimental contexts tested

in Singh et al.’s study did not seem to favor implicatures in general, even in adults, who showed surprisingly low rates of rejection in the critical implicature conditions. One possible methodological concern could be related to their use of a TVJT in description mode; uttering a disjunction in description mode is odd if the speaker has full knowledge of which disjuncts are true. Moreover, the study used static images instead of unfolding stories, which might have provided less supportive contexts for children to draw implicatures. In fact, almost half of the children (46%) failed to pass the controls in the experiment. Altogether, this led many researchers to question the reliability of the results.

Subsequent to Singh et al.’s study, however, further support for their proposal was reported in Tieu et al.’s (2017) study of French- and Japanese-acquiring children’s interpretation of disjunction. These authors tested children’s interpretation of simple (“ou” in French, “ka” in Japanese) and complex (“soit... soit” in French, “ka... ka” in Japanese) disjunctions, using a TVJT in prediction mode, in which children heard a puppet make a guess about the outcome of a scenario, and, upon seeing the outcome, had to say if the puppet had guessed correctly. The use of prediction mode, as well as animated stories, was aimed at addressing the methodological concerns associated with Singh et al.’s (2016) study, and led to very low exclusion rates. While the adult controls in both languages generally interpreted both forms of disjunction exclusively, children clustered into inclusive or conjunctive sub-groups. No significant differences were found between the two languages, nor between simple and complex disjunctions. Tieu et al. took their findings to support Singh et al.’s proposal, and the alternatives-based account of children’s difficulties with implicatures more generally.

In further corroborating work, Pagliarini et al. (2018a) tested children’s interpretation of disjunction under the quantifier “every” and observed that children accessed either literal, conjunctive, or adult-like distributive interpretations. In contrast, however, Sauerland and Yatsushiro (2018) extended Tieu et al.’s (2017) study to German-speaking children, and reported more inclusive and exclusive children than conjunctive, for both simple (“oder”) and complex (“entweder oder”) disjunctions, suggesting the need for further cross-linguistic work.

Subsequent work argued that the conjunctive behavior reported in Singh et al. (2016) and Tieu et al. (2017) reflected an experimental artifact, rather than a genuine interpretive option for children. Skordos et al. (2020) pointed out that disjunctive sentences are underinformative/trivial in a context that contains only the two mentioned disjuncts, if it is assumed that at least one object will be acted on. The authors hypothesized that children’s conjunctive behavior arises in response to pragmatic infelicity. They reported that fewer conjunctive responses were observed when there were three alternatives in the context (importantly, one more than was mentioned in the disjunctive sentence). In passing, they showed that a slight simplification of the experimental script also reduced the number of conjunctive responders. Similarly, Huang and Crain (2020) found that with three pictured alternatives in the context, Mandarin-speaking children interpreted disjunction inclusively, rather than conjunctively.

In a study investigating disjunction in both affirmative and negative

sentences in French, Cochard et al. (2023) addressed the potential infelicity identified by Skordos et al. (2020) by making the ‘neither-disjunct-true’ possibility more salient, and indeed observed a relatively high rate of conjunctive readings for affirmative sentences. They also observed that all children who were conjunctive in the affirmative case had a neither-nor reading under negation, which confirms that the children did not merely mistaken disjunction for conjunction.

In a study with English-speaking children, Jasbi and Frank (2021) used binary and ternary forced choice tasks and found no evidence of conjunctive readings (though no individual participant response patterns were reported). Crucially, their experiments made the conjunctive alternative salient, which could have facilitated the derivation of the exclusivity implicature, rather than a conjunctive reading. In a study with Mandarin-speaking children, Zhu and Chen (2026) implemented a TVJT with four objects instead of two, and tested *huò* (“or”) and *hé* (“and”) separately. Children reportedly interpreted both as inclusive disjunctions. The authors interpreted this result as an artifact of the task and context, and in a subsequent ternary task where each response corresponded to a distinct interpretation, they found that children were overwhelmingly conjunctive. Bhaumik and Jasbi (2026) identified informativity as the main source of potential experimental artifacts, and used an updated Give-X task to address this confound. They found adults to be mostly inclusive in this context, but children remained split between inclusive, exclusive and conjunctive responders.

Shimada and Goro (2021) tested Japanese children’s interpretation of disjunction scoping either above or below negation. Under negation, they confirmed the results of Goro and Akiba (2004): all children had a *neither...nor* interpretation (in contrast to Japanese adults, for whom the disjunction is a positive polarity item that must take scope above disjunction). For disjunctions that scoped above negation, however, children were, somewhat surprisingly, adult-like. Singh et al.’s account would lead us to expect at least some children to be conjunctive in this case (a disjunction scoping above negation is in an upward-entailing environment and could in principle be strengthened to a conjunction), but no child accessed this interpretation. Instead, nearly all children had an adult-like *exclusive* interpretation of the disjunction, which the authors took as evidence against the implicature account. Note, however, that the experiment also included control items with conjunction (...*mo*...*mo*), and that Singh et al.’s account precisely predicts that children who have access to the conjunctive alternative cannot derive a conjunctive interpretation of disjunction.

By contrast, in an experiment with Romanian children, Bleotu et al. (2025b) found that simply including items with a conjunction was not sufficient to trigger exclusive readings. Exclusive readings were only facilitated when participants were exposed to conjunction within a Question Under Discussion that made the conjunction relevant. It is important to note, however, that participants in this study exhibited a rather low baseline rate of conjunctive readings (mostly preferring inclusive readings), presumably because, following Huang and Crain (2020) and Skordos et al. (2020), the target trials always included four objects pictured in the con-

text, while Shimada and Goro’s contexts contained two. Bhaumik (2025, 2026) subsequently presented converging results from Bengali-speaking children indicating that relevance is more important than contextual mention of the alternative. The experimental design doesn’t allow us to assess individual interpretations, but following a conjunctive QUD, 5-year-olds were as a group indistinguishable from 9-year-olds and quite similar to adults (66% vs. 69% and 85% target responses, respectively).

In a subsequent study, Bleotu et al. (2025c) compared children’s interpretation of different disjunctive markers in a two-object context and a four-object context, the idea being that if conjunctive interpretations were an experimental artifact associated with the underinformativity of mentioning all relevant alternatives, one should observe fewer conjunctive readings in the four-object context. While increasing the number of objects in the context did decrease conjunctive behavior for both simple “sau” and complex “sau...sau” disjunctions, the rate of conjunctive readings remained significantly above chance.

Finally, returning to FC inferences (which motivated Singh et al.’s implicature account of children’s conjunctive interpretations), Cochard et al. (2025) raise the possibility that children who appeared to compute FC inferences in earlier studies may in fact have simply been displaying conjunctive behavior. The authors show that some French-speaking children access a stronger interpretation, whereby the disjunction is interpreted as a conjunction under the modal: “you can have the apple or the banana” would mean that you are allowed to take both fruits as a package, but not necessarily to freely choose one or the other. However, since the implicature account can capture this interpretation by assuming local strengthening within the scope of the modal (as Singh et al. 2016 and Pagliarini et al. 2018a do for disjunction under “every”), these results do not decisively advocate for one or another theoretical account. Finally, Bleotu et al. (2024) show that Romanian-speaking children who derive this modalized conjunctive reading also tend to be conjunctive with plain disjunctions.

In sum, this period saw a resurgence in the interest in conjunctive interpretations, this time from a pragmatic perspective, which led to formal proposals that could capture children’s conjunctive behavior. On the other hand, such formal proposals have had to contend with counterarguments that explain conjunctive readings away as an experimental artifact. This debate has led to increasingly targeted and careful investigations of the different factors affecting conjunctive behavior.

6 Discussion

As we have seen, the debate surrounding the source of conjunctive interpretations has not yet been settled. Having reviewed the previous literature in some detail, here we would like to assess whether past findings might be brought to bear on the competing accounts, and discuss some general issues arising from the previous literature.

6.1 Disjunctive reasoning

Let us first examine children’s success on Task 4 from Braine and Romain (1981), which was considered a main challenge to the idea that children interpret disjunction as conjunction.⁸ This task tested the three different reasoning schema in (3) (instantiated with various numbers of problems).

- (3) a. $p \vee q, \neg p \vdash q$
- b. $p \rightarrow r, q \rightarrow r, p \vee q \vdash r$
- c. $(p \vee q) \rightarrow r, p \vdash r$

In (3a) (*disjunctive syllogism*), the second premise $\neg p$ (“not p ”) would directly contradict the first premise if the latter were interpreted conjunctively. Assuming that implicatures are cancelable, conjunctive children should therefore default back to the inclusive reading upon hearing the second premise. Alternatively, they could stick with a conjunctive interpretation and dismiss the second premise. In either case, the conclusion q would follow, so conjunctive children would be predicted to succeed. In (3b) (*disjunction elimination*), the conclusion follows no matter how the disjunction is interpreted (inclusive, exclusive, or conjunctive). Finally, in (3c), disjunction appears in the antecedent of a conditional, where implicatures are blocked. Children, like adults, then adopt the standard inclusive interpretation of disjunction, allowing them to solve (3c). In short, the finding that children succeed with all reasoning schema in (3) does not provide compelling evidence against the idea that children can interpret disjunction as a conjunction by deriving an implicature.

6.2 Sources of conjunctive behavior

The implicature account offers an explanation for children’s conjunctive interpretations under the assumption that this behavior corresponds to a genuine interpretation of disjunction. The main alternative approach is to assume that this behavior does not in fact reflect a genuine interpretation, but is rather an artifact of the tasks used to test children.⁹ Both explanations are in principle compatible with all of the previous findings of conjunctive behavior, so the debate between the two revolves primarily around which one is more parsimonious and has the best explanatory power.

The main limitation of the experimental artifact explanation is that it does not predict conjunctive interpretations specifically: even assuming that children feel the need to deviate from the inclusive interpretation of

⁸Singh et al. (2016) cite Braine and Romain’s study, but only refer to Task 3, which focused on truth conditions and did reveal conjunctive behavior.

⁹There are other alternatives which assume that children do access a genuine conjunctive interpretation. Sauerland and Yatsushiro (2018) propose an ambiguity account with an associated disambiguation mechanism, but ultimately dismiss it. A similar account assumes that children unfamiliar with a connective will default to a conjunctive interpretation (Jasbi and Frank, 2021; Bleotu et al., 2025a). Klochowiec et al. (2026) propose an alternative account based on hypothesized cognitive biases, which also predicts conjunctive interpretations in children.

disjunctive statements when they are too weak, it is nevertheless remarkable that they often seem to do so by adopting a conjunctive behavior, especially when other options exist. For instance, they could ignore one of the disjuncts and respond in line with just “A” or just “B”, or they could arrive at an exclusive interpretation “A or B but not both” through other means. Huang and Crain (2020) admit this limitation, while Skordos et al. (2020) explicitly argue that conjunctive behavior is not frequent enough to call for an explanation. Our review of the literature, however, suggests that its existence is more than anecdotal, and in particular much more frequent than exclusive behavior—the dominant interpretation in adults.

On the other hand, the current empirical landscape makes it clear that the rate of conjunctive behavior can vary vastly depending on features of the task, the language under study, and the age of the child participants. If conjunctive behavior is just an experimental artifact, this variation is expected and does not call for a particular explanation. For the implicature account on the other hand, this variation implies underlying variation in interpretation, which ought to be explained. Several mechanisms could modulate the rate of conjunctive behavior (e.g., salience of a conjunctive alternative, question under discussion), but Skordos et al. (2020) point out that capturing the two vs. three objects contrast would require very specific assumptions about how listeners disambiguate between parses. Even in adults, predicting the rate at which various candidate parses are selected during disambiguation has been a long-standing challenge for the exhaustivity approach to implicatures. In the absence of such a principle, the account is so flexible that it borders on unfalsifiability. Categorical rules (e.g., Singh 2025) cannot capture graded effects of context, and the best solution to date relies on probabilistic game-theoretic approaches (Franke and Bergen, 2020; Champollion et al., 2026). Such game-theoretic solutions have not yet been applied to acquisition, but they offer the most promising avenue to complement the implicature account, with quantitative predictions about the effect of various contextual factors on the rate at which different interpretations are selected.

In short, the implicature account should be favored if (i) conjunctive behavior is common enough to call for a specific explanation, and (ii) a unified explanation of the observed variation in the rate of conjunctive responses can be put forward without making the account over-parametrized and unfalsifiable. If these conditions are not met, conjunctive behavior may be better rationalized on a case-by-case basis as the result of various experimental artifacts. Our review suggests that (i) is met, and (ii) is probably within reach, but has not yet been achieved.

6.3 Disjunction in the input

Children’s and adults’ production of disjunction has not received as much attention in the literature compared to the comprehension of disjunction. This relates to a particularly important developmental question, namely what kinds of evidence relating to disjunction children receive in their input.

Morris (2008) reports based on a corpus study of 2–5-year-old children’s production and their parental input that “or” is about 13 times

less frequent than “and” in both, and that children do not produce “or” until at least 3 year of age. Morris also reports that both children and adults tend to use disjunction with incompatible disjuncts, i.e. two disjuncts that cannot hold simultaneously (about 75% for adults and 90% for children), in line with Neimark (1970); Johansson and Sjölin (1975); Hatano and Suga (1977). Disjunction was also produced most frequently in questions (about 90% for both children and adults). The remaining uses corresponded to FC uses or *or else* interpretations (usually with imperatives).

Jasbi et al. (2024) also report a corpus study of child and parental productions of disjunction, taking into account various factors that might affect the intended interpretation. The authors confirm that disjunction most often occurs in questions, and often with logically incompatible disjuncts which “force” an exclusive interpretation. When the disjuncts are compatible, however, both inclusive and exclusive interpretations are possible in questions, and intonation (rise vs. rise-fall) disambiguates between the two. In declaratives, the different interpretations are roughly equally likely. This is true even when focusing on non-modal sentences, which receive about 25% conjunctive interpretations, 30% exclusive, and 25% inclusive interpretations (with the rest corresponding to possible interpretations associated with binary connectives). Interestingly, the authors also trained a computational model to distinguish the different interpretations based on the various cues they measured. When the two disjuncts were compatible, the resulting model always assigned a conjunctive interpretation to “or” in affirmative declarative sentences (aligning with the heuristic proposed by Hatano and Suga 1977).

Overall, these studies indicate that the input clearly separates “and” from “or”, to a point where it might even be difficult for children to realize that these two words can compete (in contrast to other Horn scales such as ⟨some, all⟩, where the distributions of the two terms are more likely to overlap). Given most occurrences of disjunction in the input involve questions, the input actually does not offer much evidence that disjunctions should be exclusive: labeling questions as involving “exclusive” or “inclusive” interpretations is a stretch, since questions do not have truth conditions to begin with.¹⁰ Besides, the majority of cases involve incompatible disjuncts, which block a conjunctive interpretation but neutralize the distinction between inclusive and exclusive.

Acquisition studies typically focus on declarative sentences with compatible disjuncts, and in this case the input is largely uninformative. If anything, we know from cross-linguistic studies of negative declarative sentences that in many languages children’s preferred interpretation does not align with that of adults. The fact that children’s inclusive interpretation in such cases matches the input in English therefore seems purely incidental. A potentially fruitful avenue of research would be to explore how children interpret disjunction in questions, and to consider more care-

¹⁰Alternative questions are often assumed to presuppose the exclusive disjunction, which is very different from conveying it. Children seem to be more exclusive than adults here (Ariel et al., 2024; Yan, 2026), presumably because they are less likely to accommodate presuppositions.

fully how this could constrain the semantics they assign to disjunction and the pragmatic alternatives they associate with it.

7 Conclusions and outlook

What we have seen in this brief overview is that experimental studies from different eras, using different methodologies, in different theoretical contexts, against a diverse set of research questions and hypotheses, and across different languages, have converged on certain findings (see Table 2 for a condensed summary). The general empirical consensus seems to be the following: while adults interpret disjunction logically (inclusively) or exclusively, children are less likely to interpret disjunction exclusively, instead favoring inclusive or conjunctive interpretations.

The puzzling conjunctive behavior has been subject to scrutiny since it was first observed. Once thought to be the predominant pattern of interpretation, its importance has been downplayed over time. It is clearly present in some tasks, but negligible in others. Studies focusing on negative environments and those testing children’s reasoning abilities have convincingly established that children do not simply confuse “or” and “and”. However, a comprehensive account of the very diverse set of data collected over more than 60 years remains challenging.

Singh et al. (2016) have proposed to explain conjunctive interpretations in children as an implicature derived by accessing a set of alternatives different from that of adults. While promising, the account has sparked intense debate in the literature. The main point of contention is that recent TVJTs mainly detect conjunctive interpretations in contexts with only two objects, which make the inclusive interpretation near trivial.

This could suggest that the conjunctive interpretation is simply an experimental artifact. But if so, additional assumptions are needed to explain why children dissatisfied with a trivial inclusive interpretation systematically strengthen it to a conjunction. Conversely, while an implicature account would predict that the context and question under discussion can affect the rate of implicature calculation, deriving precise predictions would require fleshing out the exact mechanisms at play.

Finally, we turn to the question of why we began our review so far back in the literature. Findings from studies conducted more than half a century ago cannot receive the same weight as recent work given substantial methodological advances—most notably the shift from Give-X tasks to the TVJT. Nevertheless, these early studies raised important questions about the acquisition of disjunction that have received comparatively little attention in subsequent research, and some of their findings still merit proper explanations. First, the experimental pragmatics approach has mostly focused on preschool-aged children, and has not properly explored the developmental trajectory children take towards adult-like interpretations. Older studies suggested that this might not happen until the late teenage years, but this should be confirmed with the modern methodologies now at our disposal. On top of this, the implicature account potentially predicts a transition from conjunctive to exclusive interpretations. Because these two interpretations directly contradict each other, it is difficult to

envisage a gradual shift from one to the other; studying this transition in detail would therefore be quite informative (and could in fact provide the best shot at falsifying the account).

Second, almost all studies of children’s interpretation of disjunction have focused on plain declarative sentences, but early free production experiments and recent corpus studies indicate that disjunction is much more prevalent in questions or under modals. This of course doesn’t mean that studying declaratives is uninteresting, but the explanations proposed for children’s behavior in such contexts need to also take into account the fact that most of the evidence children receive for disjunction comes from alternative questions.

Other important questions deserving of more attention include the effect of overlapping vs. non-overlapping disjuncts, effects of modality, and children’s use of “and” and “or” in production, where these expressions frequently serve functions beyond simple logical conjunction and disjunction.

It is our hope that against the historical context provided here, the present review of these findings will inform more targeted, future investigations into the acquisition of disjunction.

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A Derivation of the conjunctive reading

The implicature account can derive a conjunctive reading of disjunction provided the conjunction does not enter into the set of alternatives under consideration. The core idea is to consider not only the literal sense of the alternatives, but also their respective implicatures. Several versions of this idea exist, using stacked exhaustivity operators (Fox, 2007), a modified operator which can both exclude and *include* alternatives (Bar-Lev and Fox, 2020), recursive game-theoretic reasoning (Franke, 2011) or a mix of exhaustification and game-theory (Champollion et al., 2026). Our goal here is to informally sketch how this idea can work; we refer interested readers to the cited papers for more formal derivations.

Imagine we want to know what Sarah ate, and the speaker utters (4). An adult would typically consider the alternatives in (5).

- | | | |
|-----|-------------------------------------|--------------|
| (4) | Sarah ate an apple or a banana. | $A \vee B$ |
| (5) | a. Sarah ate an apple. | A |
| | b. Sarah ate a banana. | B |
| | c. Sarah ate an apple and a banana. | $A \wedge B$ |

By the maxims of quantity and quality, we would conclude that the speaker does not take any of the alternatives in (5) to be true. By further assuming that the speaker is sufficiently informed (the *epistemic step*), we could conclude that (5c) must be false. We cannot assume that the speaker is well-informed about (5a) or (5b), because negating one would immediately affirm the other (by the disjunctive syllogism), contradicting the first inferences above. An adult will thus end up with an exhaustive interpretation (the speaker takes A or B to be true, but not both) and ignorance inferences (the speaker doesn't know whether A is true, nor whether B is true).

In order to reach a conjunctive interpretation, the first step is to drop (5c) from the set of alternatives under consideration (a common assumption to explain why children do not access exclusive readings of disjunction). In the usual Gricean theory, however, stopping here would only get us ignorance inferences. The next step is to consider not just the literal meaning of (5a) and (5b), but also the implicatures they would each give rise to: if B is salient, (5a) would typically be interpreted as “Sarah ate an apple, but not a banana”. The child pragmatic interpreter would therefore conclude that if the speaker thought that Sarah ate only the apple, she would have uttered (5a), and if she thought Sarah had eaten only the banana, she would have uttered (5b). In other words, it's not the case that Sarah ate only one of the fruits. Since the speaker uttered (4), Sarah must have eaten at least one of the fruits, and so we can exclude the situation where Sarah ate neither the apple nor the banana. This leaves only one possibility: Sarah must have eaten both the apple and the banana.

Note that considering strengthened alternatives together with (5c) would not have any effect for a plain disjunction, but it can explain how adults derive free choice.